

Hide menu

Generalized Additive Models: Basics

Reading: Required: Generalized additive models for data science 10 min

Video: Mockingbird: Generalized Additive Models 17 min

Video: Generalized Additive Models in R 16 min

Lab: Generalized Additive Models in R 1h

Quiz: Generalized Additive Models: Basics Submitted

Generalized Additive Models: Inference and Data Analysis

Assignments: GAMs

● Congratulations! You passed!

Grade received 100%

Latest Submission

To pass 80% or higher

Go to next item

Generalized Additive Models: Basics

Review Learning Objectives

1. When compared to some common machine learning techniques, such as random forests, generalized additive models have the advantage of clearly showing the contribution of each predictor to the response. 3 / 3 points

Ⓐ True

Ⓑ False

Ⓒ Correct

● Submit your assignment

Due Jun 16, 11:59 PM IST

Try again

● Receive grade

To Pass: 80% or higher

Your grade 100%

View Feedback

2. Generalized additive models can be thought of as a way to estimate nonlinear relationships between a response and several predictors simultaneously. 3 / 3 points

Ⓐ True

Ⓑ False

Ⓒ Correct

👍 Like

👎 Dislike

🚩 Report an issue

3. Generalized additive models strike a nice balance between the interpretable, yet biased, linear model, and the extremely flexible, "black box" machine learning algorithms. 3 / 3 points

Ⓐ True

Ⓑ False

Ⓒ Correct

4. Which of the following are additive models? 10 / 10 points

☐ $f(x_1, x_2, x_3) = \beta_0 x_1 + \beta_1 x_1 + \cos(\pi x_2) + \beta_3 x_2^2 + x_1 x_2 + \sin^2(x_3)$

☒ $f(x_1, x_2) = \pi + \beta_1 e^{x_1} - 5^{x_2}$

Ⓒ Correct

☒ $f(x_1, x_2, x_3) = \beta_0 x_1 + \beta_1 x_1 + \cos(\pi x_2) + \beta_3 x_2^2 + \sin^2(x_3)$

Ⓓ Correct

☒ $f(x_1, x_2, x_3) = \beta_0 + \beta_1 \log(x_1 x_3^5) - \beta_2 5^{x_2}$

Ⓔ Correct

Log laws!

☐ $f(x_1, x_2, x_3) = \beta_0 + \beta_1 e^{x_1/x_2} - \beta_2 5^{x_2}$

5. Additive models will work well when strong interactions between predictors exist. 3 / 3 points

Ⓐ True

Ⓑ False

Ⓒ Correct

6. Generalized additive models have trouble incorporating non-normal (e.g., binomial) responses. 3 / 3 points

Ⓐ True

Ⓑ False

Ⓒ Correct

7. Generalized additive models are typically more biased than standard linear regression models. 3 / 3 points

Ⓐ True

Ⓑ False

Ⓒ Correct

8. Suppose that a response y is related nonlinearly to a (continuous) predictor x_1 , linearly to a (continuous) predictor x_2 , and linearly to a three-level factor x_3 . Then: 4 / 4 points

Family: gaussian
Link function: identity

Formula:
 $y \sim s(x_1) + x_2 + x_3$

Parametric coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 1.49974 0.36072 4.158 9.25e-05 ***
x2 2.81405 0.12020 23.412 < 2e-16 ***
x3B 1.02847 0.02579 39.875 < 2e-16 ***
x3C 1.94796 0.02498 77.973 < 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
edf Ref. df F p-value
s(x1) 8.283 8.053 600.9 <2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.994 Deviance explained = 99.5%
GCV = 0.0091693 Scale est. = 0.0077614 n = 80

☒ For a one-unit increase in x_2 , the mean change in y is approximately 2.81, adjusting for other predictors.

Ⓒ Correct

☐ This model includes interaction terms.

☐ For a one-unit increase in x_1 , the mean change in y is approximately 8.28, adjusting for other predictors.

☒ The mean change in y for one-unit increase in x_1 , adjusting for other predictors, depends on the value of x_1 .

Ⓓ Correct

https://www.coursera.org/learn/generalized-linear-models-and-nonparametric-regression/exam/q76e/generalized-additive-models-basics/attempt/7redirectToCover=true

1/1